1)

a) False (see p. 4 in the lecture notes)

b) True (see p. 11 in the lecture notes)

c) True (see Theorem 3.10 b) in the lecture notes)

d) False (We have to specify the distribution of the first variable, X_1, and the second variable, X_2, independently. For each variable we need 5 numbers. The product distribution is then specified by 25 numbers.)

e) False (There is a path from A to D that does not go through B.)

f) False (Statistical models can be wild, giving rise to many minimisers of the KL-divergence. If we assume the statistical model to be an exponential family, however, existence and uniqueness of the minimiser can be shown.)

2)

a) There exist $\mu_A, \mu_B, \mu_C \in \mathbb{R}$ and $\sigma \in (0, \infty)$ such that the age of a player from team $j \in \{A, B, C\}$ is $\text{Normal}(\mu_j, \sigma^2)$-distributed. The ages of different players are independent.

b) For $j \in \{A, B, C\}$ let $\hat{\mu}_j$ be the sample mean of team j. We have

$$\hat{\mu}_A = 26.9965$$

$$\hat{\mu}_B = 26.9965 + 2.5845 = 29.5810$$

$$\hat{\mu}_C = 26.9965 - 0.9060 = 26.0905.$$

The estimate $\hat{\sigma}$ for σ is

$$\hat{\sigma} = 3.447.$$

c) We test

$$H_0: \mu_A = \mu_B = \mu_C$$

against

$$H_1: \mu_i \neq \mu_j \text{ for some } i, j \in \{A, B, C\}.$$

Since the p-value 0.001812 is less than 0.05, we can reject H_0 at level 0.05, i.e. there is a difference between the teams.

d) We test

$H_0^{(1)}: \mu_A = \mu_B$ against $H_1^{(1)}: \mu_A \neq \mu_B$,

$H_0^{(2)}: \mu_A = \mu_C$ against $H_1^{(2)}: \mu_A \neq \mu_C$,

$H_0^{(3)}: \mu_B = \mu_C$ against $H_1^{(3)}: \mu_B \neq \mu_C$.

Since their p-values are below 0.05, we can reject $H_0^{(1)}$ and $H_0^{(3)}$ at level 0.05, while we cannot reject $H_0^{(2)}$.

We can conclude that there is a difference between teams A and B and teams B and C. In fact, the players of B are older than those of A and C. The relation between A and C is not clear.

3)

a) A least square estimator $(\hat{\alpha}_0, \hat{\alpha}_1)^T$ for $(\alpha_0, \alpha_1)^T$ is a solution of

$$\min_{\tilde{\alpha}_0, \tilde{\alpha}_1 \in \mathbb{R}} \quad (10-\tilde{\alpha}_0)^2 + (7.5-\tilde{\alpha}_0 e^{-\tilde{\alpha}_1})^2 + (5-\tilde{\alpha}_0 e^{-2\tilde{\alpha}_1})^2 + (3-\tilde{\alpha}_0 e^{-3\tilde{\alpha}_1})^2.$$

b) We have that

$$P(\text{the part is destroyed}) = \frac{\exp(\beta_0+\beta_1 t)}{1+\exp(\beta_0+\beta_1 t)}.$$

c) We know that for

$H_0: \langle a, \gamma \rangle = c_0$ against $H_1: \langle a, \gamma \rangle \neq c_0$,

$$\varphi_{\neq}(Y) = \begin{cases} \text{cannot reject}, & c_0 \in I^{\neq}_{a,\hat{\gamma}}, \\ \text{reject}, & c_0 \notin I^{\neq}_{a,\hat{\gamma}}, \end{cases}$$

with

$$I^{\neq}_{a,\hat{\gamma}} = \left(\langle a, \hat{\gamma} \rangle - t_{n-1-k, 1-\frac{\alpha}{2}} \hat{\sigma} \sqrt{a^T (A^T A)^{-1} a}, \quad \langle a, \hat{\gamma} \rangle + t_{n-1-k, 1-\frac{\alpha}{2}} \hat{\sigma} \sqrt{a^T (A^T A)^{-1} a} \right)$$

is a level-α test. We have $k=2$, $a=(0,1,-1)^T$ and $c_0=0$ and obtain

$$a^T (A^T A)^{-1} a = (0,1,-1) \begin{pmatrix} n & & \\ & 2 & \\ & & 2 \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$= (0,1,-1) \begin{pmatrix} \frac{1}{n} & & \\ & \frac{1}{2} & \\ & & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2} + \frac{1}{2} = 1.$$

Thus,

$$I^{\neq}_{a,\hat{\gamma}} = \left(\hat{\gamma}_1 - \hat{\gamma}_2 - t_{n-3, 1-\frac{\alpha}{2}} \hat{\sigma}, \ \hat{\gamma}_1 - \hat{\gamma}_2 + t_{n-3, 1-\frac{\alpha}{2}} \hat{\sigma} \right),$$

which completes the proof.

4)

1. $\langle a,b\rangle_\mu = \langle (2,4,2),(-5,2,1)\rangle_\mu$

$$= \frac{1}{1}\cdot 2\cdot(-5) + \frac{1}{2}\cdot 4\cdot 2 + \frac{1}{3}\cdot 2\cdot 1$$

$$= -10 + 4 + \frac{2}{3} = -\frac{30}{3} + \frac{12}{3} + \frac{2}{3}$$

$$= -\frac{16}{3} \approx -5{,}33$$

2. $\|a\|_\mu = \|(2,4,2)\| = \left(\frac{1}{1}\cdot 4 + \frac{1}{2}\cdot 16 + \frac{1}{3}\cdot 4\right)^{1/2}$

$$= \left(\frac{12}{3} + \frac{24}{3} + \frac{4}{3}\right)^{\frac{1}{2}} = \sqrt{\frac{40}{3}} = 2\sqrt{\frac{10}{3}} \approx 3{,}65$$

$\|b\|_\mu = \|(-5,2,1)\| = \left(\frac{1}{1}\cdot 25 + \frac{1}{2}\cdot 4 + \frac{1}{3}\cdot 1\right)^{1/2}$

$$= \left(\frac{75}{3} + \frac{6}{3} + \frac{1}{3}\right)^{1/2} = \sqrt{\frac{82}{3}} \approx 5{,}23$$

3. $$\cos\alpha = \frac{\langle a,b\rangle_\mu}{\|a\|_\mu\cdot\|b\|_\mu} = \frac{-16/3}{2\sqrt{\frac{10}{3}}\sqrt{\frac{82}{3}}} = -\frac{8}{\sqrt{10}\sqrt{82}}$$

$$= -\frac{4}{\sqrt{5}\sqrt{41}} \approx -0{,}28$$

4. $$\alpha = \cos^{-1}\left(-\frac{4}{\sqrt{5}\sqrt{41}}\right) \approx 106{,}22^\circ$$

5)

1. Hellinger distance:

$$d^H(\mu_1,\mu_2) = 2\left(\left(\sqrt{\tfrac{1}{6}}-\sqrt{\tfrac{1}{2}}\right)^2+\left(\sqrt{\tfrac{1}{3}}-\sqrt{\tfrac{1}{3}}\right)^2+\left(\sqrt{\tfrac{1}{2}}-\sqrt{\tfrac{1}{6}}\right)^2\right)^{\frac{1}{2}}$$

$$= 2\sqrt{2\left(\sqrt{\tfrac{1}{2}}-\sqrt{\tfrac{1}{6}}\right)^2} = 2\cdot\sqrt{2}\left(\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{6}}\right)$$

$$= 2\left(1-\frac{1}{\sqrt{3}}\right) \approx 0{,}85$$

2. Bhattacharya coefficient:

$$\sum_{i\in I}\sqrt{\mu_{1,i}\cdot\mu_{2,i}} = \sqrt{\tfrac{1}{6}}\cdot\sqrt{\tfrac{1}{2}}+\sqrt{\tfrac{1}{3}}\cdot\sqrt{\tfrac{1}{3}}+\sqrt{\tfrac{1}{2}}\cdot\sqrt{\tfrac{1}{6}}$$

$$= 2\sqrt{\tfrac{1}{6}}\cdot\sqrt{\tfrac{1}{2}}+\frac{1}{3} = \frac{1}{\sqrt{3}}+\frac{1}{3} \approx 0{,}91$$

3. Fisher-Rao distance:

$$d^{FR}(\mu_1,\mu_2) = 2\cdot\cos^{-1}\left(\frac{1}{\sqrt{3}}+\frac{1}{3}\right) \approx 0{,}85$$

4. Kullback-Leibler divergence $D_{KL}(\mu_1\|\mu_2)$:

$$D_{KL}(\mu_1\|\mu_2) = \frac{1}{6}\ln\frac{1/6}{1/2}+\frac{1}{3}\ln\frac{1/3}{1/3}+\frac{1}{2}\ln\frac{1/2}{1/6}$$

$$= \frac{1}{6}\ln\frac{1}{3}+\frac{1}{2}\ln 3 = \left(\frac{1}{2}-\frac{1}{6}\right)\ln 3$$

$$= \frac{1}{3}\ln 3 \approx 0{,}37$$